

AADITYA SATTAWAN

EDUCATION

Indian Institute of Technology (BHU), Varanasi	2023 - 2028
<i>Integrated Dual Degree in Mechanical Engineering</i>	7.25
Bharatiya Vidya Bhavan Vidyashram	2023
<i>CBSE</i>	87.80%
Bharatiya Vidya Bhavan Vidyashram	2021
<i>CBSE</i>	90.20%

SKILLS

- **Programming Languages:** Python, SQL, R, C++, Java.
- **ML & AI Libraries:** Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, XGBoost.
- **Data Analysis & Visualisation:** Matplotlib, Seaborn, Plotly, Tableau, Power BI, Excel.
- **Databases & Tools:** MySQL, PostgreSQL, Git, GitHub, Docker, Jupyter Notebook, Google Colab.

INTERNSHIPS

AI & Data Science Intern XYlofyAI	Jun 2026 - Jul 2026
<i>Developed production-quality machine learning solutions and business analytics workflows.</i>	Jaipur
• Developed ML pipelines for house price prediction and employee attrition analysis. • Performed preprocessing, EDA, feature engineering, and trained multiple ML models. • Created visualisations and executive reports with actionable business insights.	

PROJECTS

InterviewIQ	
<i>AI-powered interview preparation platform for resume-job analysis and personalised interview planning</i>	May 26 - Jun 26
• Built an AI-driven interview platform using Google Gemini for personalised interview reports. • Engineered async job processing with BullMQ and Redis for scalable AI task execution. • Implemented multi-model failover logic to ensure reliable AI generation under rate limits. • Optimised frontend state management using TanStack Query for polling and caching.	
SyncSpace-AI	
<i>AI-powered 3D workspace planner enabling workstation design through natural language commands.</i>	Jun 26 - Jul 26
• Built a real-time AI workspace planner using Next.js, React Three Fiber, and Socket.IO. • Designed an AABB-based layout engine for collision-free and ergonomic workspace generation. • Integrated Google Gemini with intent classification for intelligent task routing. • Reduced latency using heuristic optimisation for budget and layout modification requests.	
DogStudio	
<i>High-performance 3D web experience featuring interactive WebGL rendering and scroll-driven animations.</i>	May 26 - May 26
• Engineered an immersive 3D web experience using React Three Fiber, Three.js, and GSAP. • Implemented scroll-linked transformations and custom GLSL shader-based texture transitions. • Optimised rendering performance using Draco-compressed assets and ref-based animation updates. • Integrated advanced WebGL effects with hardware-accelerated animations while maintaining 60 FPS.	

POSITIONS OF RESPONSIBILITY

Captain, Aquatics Team IIT (BHU) Varanasi
• Led the institute aquatics team in training, competitions, and Inter-IIT campaigns. • Managed team operations, scheduling, athlete selection, and event coordination. • Mentored team members and promoted a culture of discipline, collaboration, and excellence.

ACHIEVEMENTS

- Codeforces Rating: **1501 (Specialist)**.
- Achieved global ranks of **1085** in Educational Codeforces Round 190 (Div.2), **1684** in Codeforces Round 1099 (Div.2), and **1244** in Codeforces Round 1103 (Div.3).
- Ranked **4th** among **200+** students at IIT (BHU) Varanasi in the **AlgoUniversity Dynamic Programming Contest**.
- Achieved a **Top 10 finish (8th rank)** among **1000+** participants in the **TechTriad 2026** Hackathon.

EXTRA CURRICULARS

- **Silver** and **Bronze** Medallist at the **38th Inter-IIT Aquatics Meet**, representing **IIT (BHU)**.
- **Bronze** Medallist at the **37th Inter-IIT Aquatics Meet**, representing **IIT (BHU)**.
- **Gold** Medallist in **Football** at the **Inter-Hostel General Championship (GC) 2023-24** at **IIT (BHU)**.